PhishGuard – Firisar Gabriel

1. Application Concept & Value Proposition

PhishGuard is a mobile app that teaches people to recognize phishing quickly. It uses short, plain-language lessons and two practice modes: a classic multiple-choice quiz and a swipe-based visual quiz with realistic email/message screenshots. Users get instant feedback, can track progress, and can learn even offline with syncing later. Admins can add or update training content.

The app serves students, new hires, and everyday users, and also fits enterprise needs. Companies can roll it out as a lightweight awareness tool across teams, while security or training leads act as admins to manage lessons and quizzes that match internal policies.

The problem is that most phishing training is rare, long, and easy to forget, while phishing tactics change often. PhishGuard solves this by making learning fast, hands-on, and always available. Users build habits through regular practice, see realistic examples, and view their progress. Organizations gain a simple, low-effort way to keep people alert and reduce risky clicks.

2. Core Data Model

User

- id: UUID — unique user
- email: String? — optional for Anonymous
- name: String? — display name
- role: Enum(USER,ADMIN) — feature gating
- photoUrl: String? — avatar
- biometricEnabled: Boolean — quick unlock
- streakDays: Integer — current day streak
- lastActiveAt: DateTime — for streak calc

Lesson

- id: UUID
- title: String — e.g., “Recognizing Suspicious Links”
- summary: String — short subtitle
- content: String — lesson body (markdown/plain)
- order: Integer — list position

LessonProgress

- id: UUID
- userId: UUID (FK User)
- lessonId: UUID (FK Lesson)
- completion: Integer (0–100) — progress bar
- lastViewedAt: DateTime
- pendingSync: Boolean — queued for server

Quiz

- id: UUID
- title: String — e.g., “Phishing Quiz #3”
- type: Enum(CLASSIC,VISUAL)
- description: String?
- order: Integer

Question (for CLASSIC)

- id: UUID
- quizId: UUID (FK Quiz)
- text: String
- explanation: String — shown after answer

Option (for CLASSIC)

- id: UUID
- questionId: UUID (FK Question)

- text: String
- isCorrect: Boolean

VisualCard (for VISUAL swipe quiz)

- id: UUID
- quizId: UUID (FK Quiz)
- imageUrl: URL — remote asset
- localPath: String? — cached image
- label: Enum(PHISHING,LEGIT) — ground truth
- explanation: String — why

QuizAttempt (covers both quiz types)

- id: UUID
- userId: UUID (FK User)
- quizId: UUID (FK Quiz)
- score: Integer
- total: Integer
- answers: JSON
- startedAt/endedAt: DateTime
- pendingSync: Boolean

UserAchievement

- id: UUID
- userId: UUID (FK User)
- type: Enum(FIRST_QUIZ,SEVEN_DAY_STREAK,PERFECT_SCORE)
- unlockedAt: DateTime
- pendingSync: Boolean

3. User Roles & Feature Breakdown:

Roles (at least two as required). The app supports a default Learner role and an Admin (Content Manager) role.

Shared features (both roles)

- Sign in with Email/Google/Anonymous; optional biometric quick-unlock.
- Access the Training Library; read lessons; resume where left off.
- Take Classic Quiz and Visual (swipe) Quiz with instant feedback.
- See basic results after a session (score and short tips).
- Offline reading & practice with automatic sync when online.

Learner — unique capabilities

- Personal progress tracking: per-lesson completion, quiz history.
- Achievements & streaks: earn simple badges (e.g., First Quiz, 7-Day Streak).
- Detailed personal results: view past attempts with explanations.

Admin (Content Manager) — unique capabilities

- Content CRUD: create/edit/publish/unpublish Lessons, Classic Questions/Options, and Visual Cards (including image upload).
- Ordering: set display order for lessons/quizzes.
- Lightweight analytics (aggregated): total users, lessons, quizzes, and top missed items to improve content (no per-user answers by default).

Access & privacy notes

- Training content is readable by any signed-in user; progress, attempts, and achievements are private to the owning Learner.
- Admins manage content and see only aggregated stats—not individual answers—keeping the model simple and privacy-respecting.

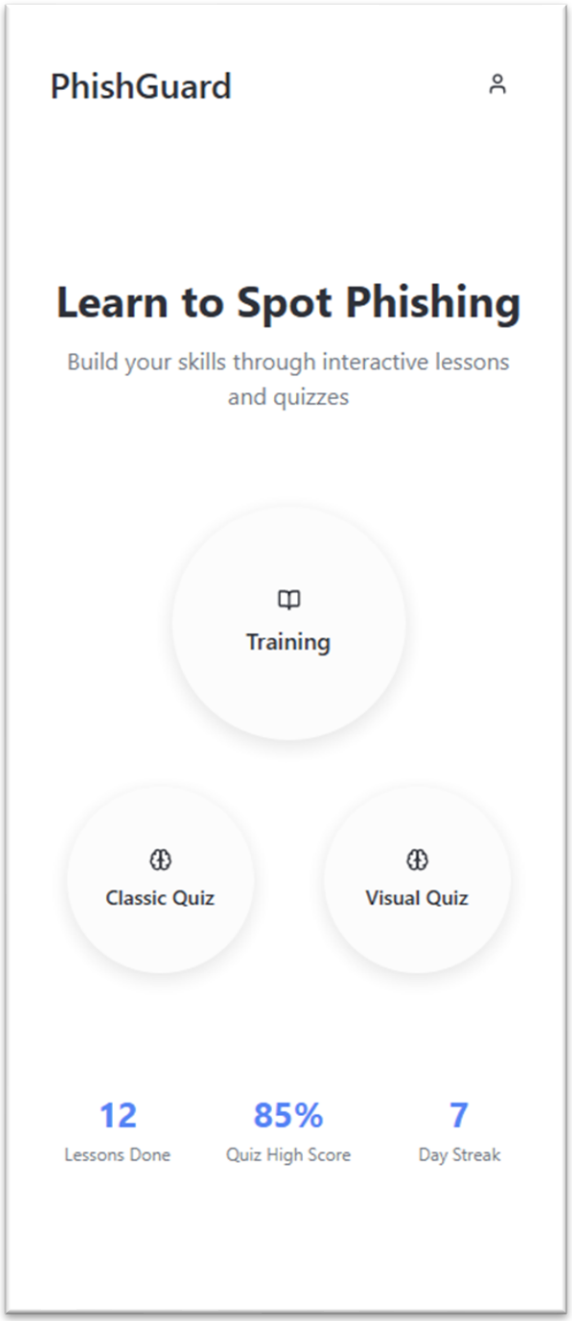
4. Offline Strategy & Synchronization

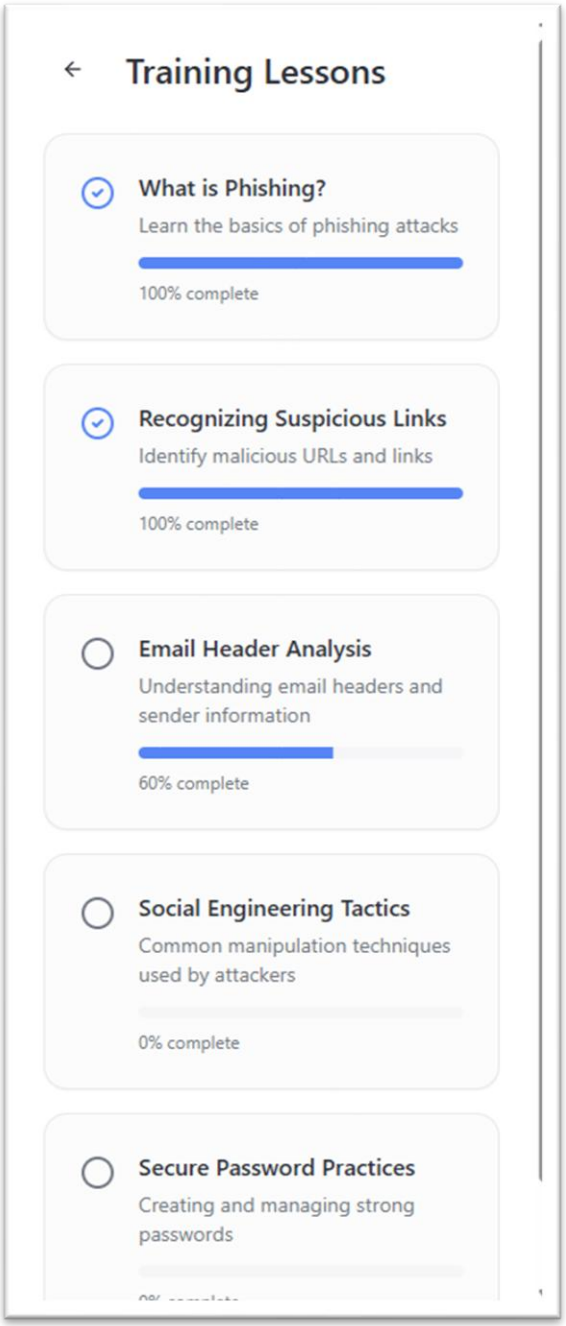
The app is offline-first: lessons and quizzes are cached in the local database so users can read and practice without internet; a subtle offline banner appears (ex: “Offline mode: recent updates may be missing. We’ll retry in the background.”), and network-dependent actions are disabled or queued until reconnect. When connectivity returns, the app refreshes content from the server and syncs locally saved progress and quiz attempts with friendly, clear feedback.

On device, we store training content (lessons, questions/options, visual cards with cached images), lesson progress, quiz attempts, and simple achievements, plus minimal sync metadata. Sensitive authentication data lives in the platform’s secure storage, and the active session uses the token issued at login for subsequent requests.

Synchronization follows a pull/push model: the app fetches updated content via the API (delta by updated timestamps) and pushes queued user data (progress, results) once online; all calls are authenticated with the login token, with retries and gentle error handling. A Repository layer decides between network and local cache so the UI remains responsive.

5. UI/UX Mockups


PhishGuard
Learn to Spot Phishing
Build your skills through interactive lessons and quizzes
Training
Classic Quiz
Visual Quiz
12
Lessons Done
85%
Quiz High Score
7
Day Streak


Training Lessons
What is Phishing?
Learn the basics of phishing attacks
100% complete
Recognizing Suspicious Links
Identify malicious URLs and links
100% complete
Email Header Analysis
Understanding email headers and sender information
60% complete
Social Engineering Tactics
Common manipulation techniques used by attackers
0% complete
Secure Password Practices
Creating and managing strong passwords

Quiz

Question 1 of 2

What is the main goal of a phishing attack?

- To entertain users with fake emails
- To steal sensitive information
- To promote legitimate products
- To test email server security

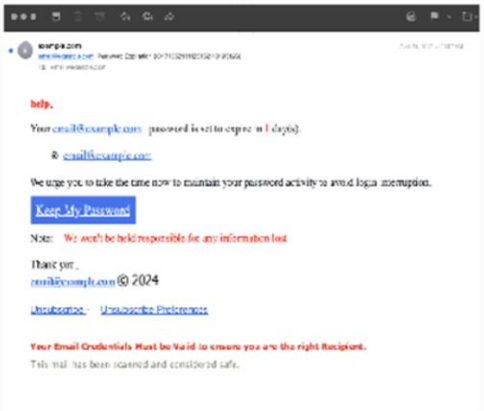
Submit Answer

Current Score **0/2**

Swipe Quiz

1 / 3

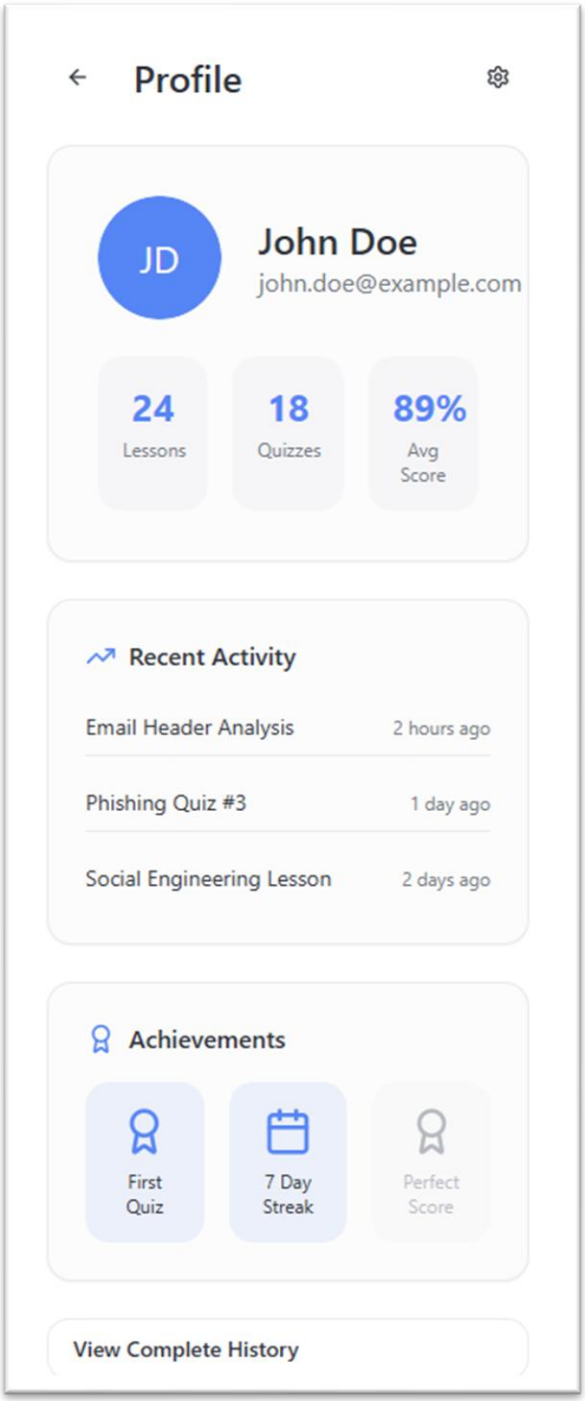
Swipe to decide if this is phishing or legitimate

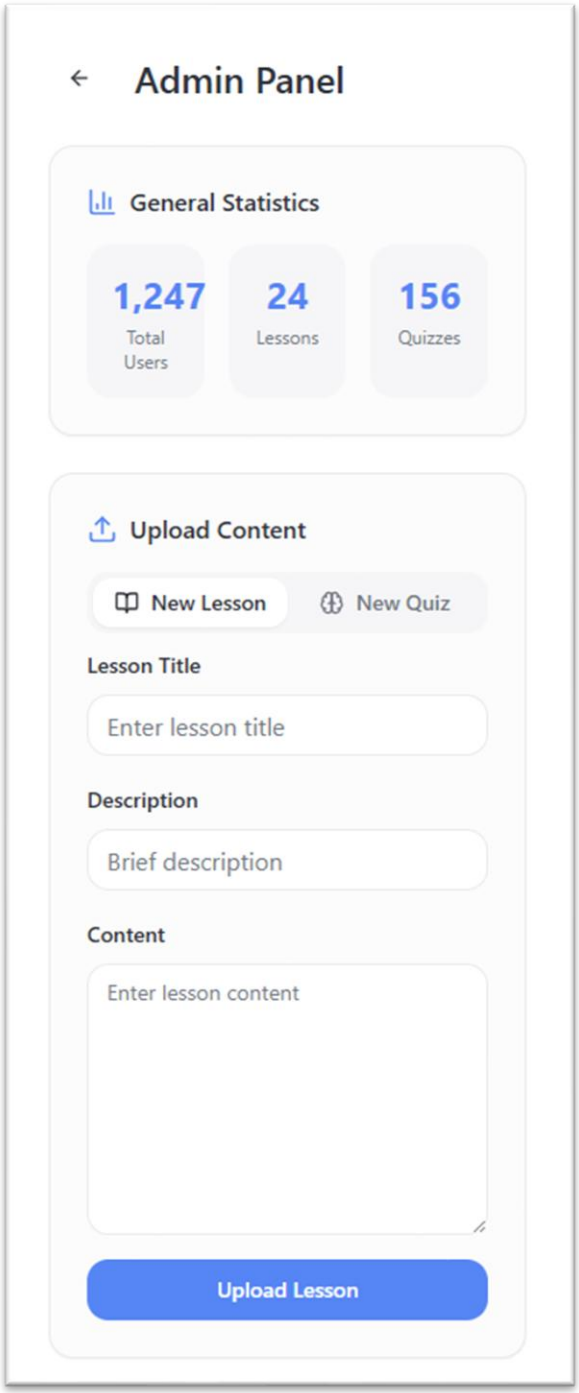


Is this phishing?

Score

0/0


Profile
JD
John Doe
john.doe@example.com
24
Lessons
18
Quizzes
89%
Avg Score
Recent Activity
Email Header Analysis
2 hours ago
Phishing Quiz #3
1 day ago
Social Engineering Lesson
2 days ago
Achievements
First Quiz
7 Day Streak
Perfect Score
View Complete History


Admin Panel
General Statistics
1,247
Total Users
24
Lessons
156
Quizzes
Upload Content
New Lesson
New Quiz
Lesson Title
Enter lesson title
Description
Brief description
Content
Enter lesson content
Upload Lesson